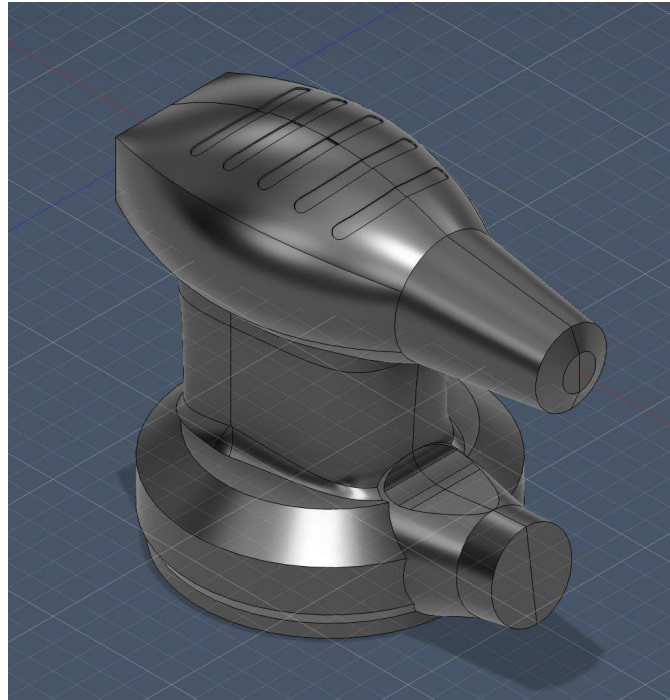


REVERSE ENGINEERING PROJECT

Final Technical Report

Inalsa Orbital (Rotary) Sander — 2.4 Amp



Fusion 360 Assembly — Inalsa Orbital Sander (Group 3-1)

Course:	ME 1212 – Mechanical Engineering Design
Section:	3
Group Number:	Group 3-1
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Abstract(DAW)

This report documents the reverse engineering of an Inalsa Orbital (Rotary) Sander (2.4 Amp) to explore the relationship between mechanical design and functional performance. The project involved complete disassembly, documentation, and measurement of the device's components. The team conducted a functional decomposition of the tool to understand how every subsystem contributes to the device's overall purpose: smoothing and leveling surfaces to a desired finish. Three specific systems were selected for in-depth analysis: the Drivetrain system, which converts motor rotation into controlled orbital motion; the Thermal/Fluid system, which manages heat dissipation and dust collection; and the Human Interfaces system, which governs all points of interaction between the user and the device. For each system, the form, fit, and function of components are described and illustrated with hand sketches, photographs, and Fusion 360 CAD models including dimensioned drawings. An improvement is proposed for each system with justification addressing cost, lifespan, and user experience. Project management is demonstrated through an updated Gantt chart.

Table of Contents

1. Introduction	3
2. Functional Decomposition Tree.....	3
3. Drivetrain System	4
3.1 Form, Fit, and Function	4
3.2 Variations / Improvements	5
4. Thermal/Fluid System	6
4.1 Form, Fit, and Function	6
4.2 Variations / Improvements	7
5. Human Interfaces System	8
5.1 Form, Fit, and Function	8
5.2 Variations / Improvements	9
6. Project Management.....	10

List of Figures

Figure 1.	Functional Decomposition Tree — Orbital Sander.....	3
Figure 2.	Fusion 360 Assembly — Full Orbital Sander.....	4
Figure 3.	Fusion 360 model of the sander backing pad.....	4
Figure 4.	Hand sketch — Motor shaft and drivetrain components.....	4
Figure 5.	Dimensioned drawing — Disk Attachment (D. Kalu).....	5
Figure 6.	Hand sketch — Fan/counterweight assembly.....	6
Figure 7.	Dimensioned drawing — Sander Housing (M. Kheshouei).....	7
Figure 8.	Project Gantt Chart — Group 3-1 (updated April 2026).....	10

1. Introduction(JDSB)

An orbital sander is a handheld power tool designed to smooth and level surfaces through the rapid, small-circle orbital motion of an abrasive pad. It is widely used in woodworking, automotive finishing, and construction to prepare surfaces for painting or coating. Reverse engineering — the process of systematically disassembling a device and analyzing its components — offers a powerful lens for understanding how a product's design choices translate into real-world function and performance. This report presents the results of a reverse engineering analysis performed on an Inalsa Orbital (Rotary) Sander (2.4 Amp) by Team 3-1 of ME 1212. The team fully disassembled the device and identified three major functional systems: the Drivetrain system, responsible for converting motor rotation into orbital sanding motion; the Thermal/Fluid system, responsible for regulating heat and collecting sanding dust; and the Human Interfaces system, responsible for all user-facing components including the grip, on/off switch, and power cord. Each system is analyzed in terms of its form, fit, and function, and a specific improvement is proposed for each.

2. Functional Decomposition Tree (DK)

The orbital sander fulfills a single overarching function: to smooth and level surfaces to a desired finish. As shown in Figure 1, this top-level function is broken down into four major sub-functions: (1) Control/Regulate inputted energy — managing electrical input to the motor and maintaining safe operating temperatures; (2) Generate Rotational Motion — converting electrical energy into rotation and controlling that rotation to minimize undesirable vibration; (3) Interlink with abrasive attachment — supporting and securing the sandpaper backing pad and enabling its controlled orbital motion; and (4) Waste Collection — capturing and directing dust and debris created during sanding. The Drivetrain system encompasses sub-functions (2) and (3). The Thermal/Fluid system encompasses sub-function (4) and thermal regulation within (1). The Human Interfaces system encompasses the user-facing aspects of (1) and (3).

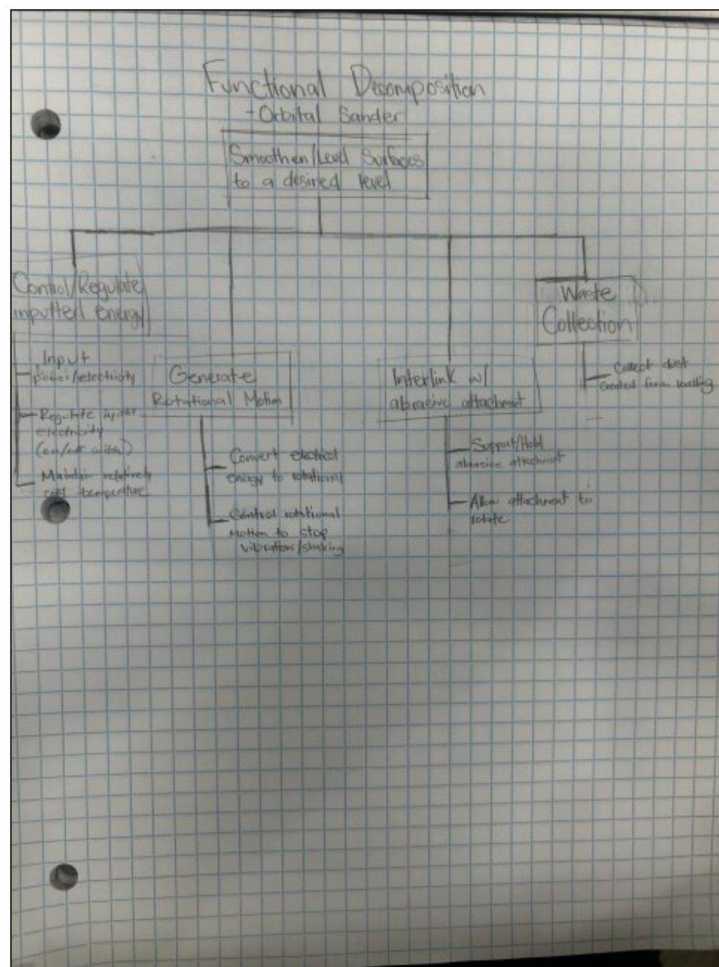


Figure 1. Functional Decomposition Tree for the Inalsa Orbital Sander (Group 3-1).

3. Drivetrain System(DK,KP)

3.1 Form, Fit, and Function

Form

The orbital sander's drivetrain is a vertical assembly centered on the steel motor shaft, which acts as the primary axis anchoring all drivetrain components during operation. As documented in the hand sketch (Figure 4), the motor shaft assembly spans approximately 107 mm in total height and consists of several distinct zones from top to bottom. At the top, a threaded screw (3.4 mm diameter, 14.8 mm long) secures the fan assembly, which consists of a fan disc (Ø52.6 mm, 3 mm thick) carrying 16 radial fan blades and a lower fan ring (Ø85 mm). Below the fan sits the rotor stack: a central shaft section (Ø24 mm, 6 mm) transitions into the rotor winding section (29.25 mm long, Ø38 mm), flanked by bearing seats at both ends (Ø36.2 mm). Beneath the rotor, the eccentric hub section (Ø25.3 mm, 16.5 mm long) is offset from the shaft centerline to generate orbital motion. The shaft terminates in a lower journal (Ø19 mm, 6 mm) that mates with the lower bearing housing. The backing pad, shown in Figure 3, is a circular disc approximately 120 mm in diameter featuring a central hub bore and a radial pattern of dust-collection through-holes.

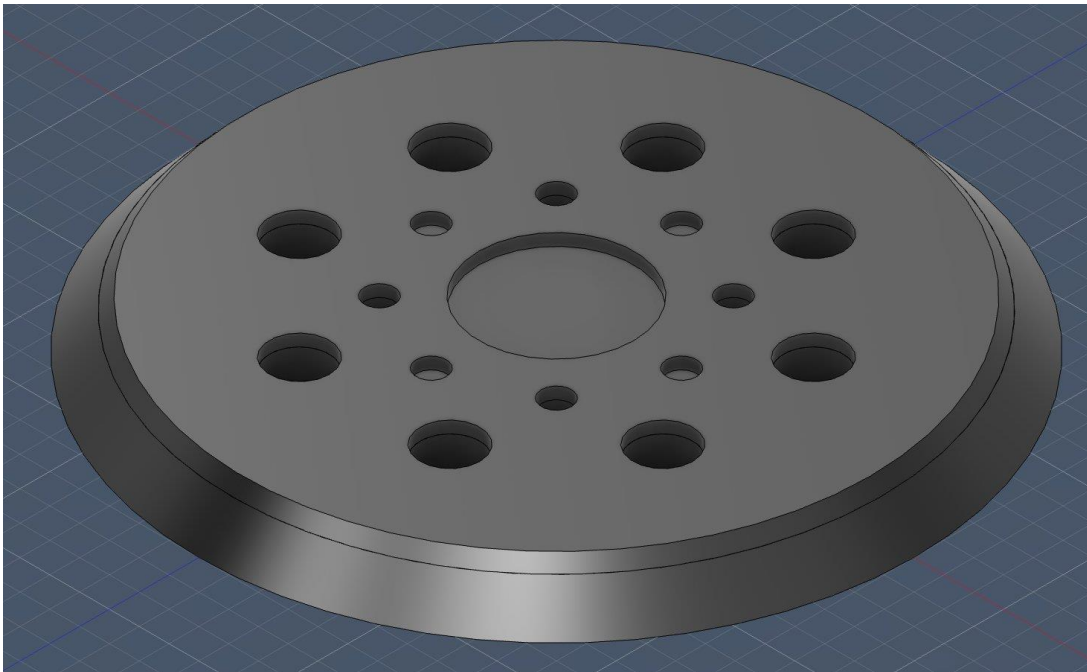
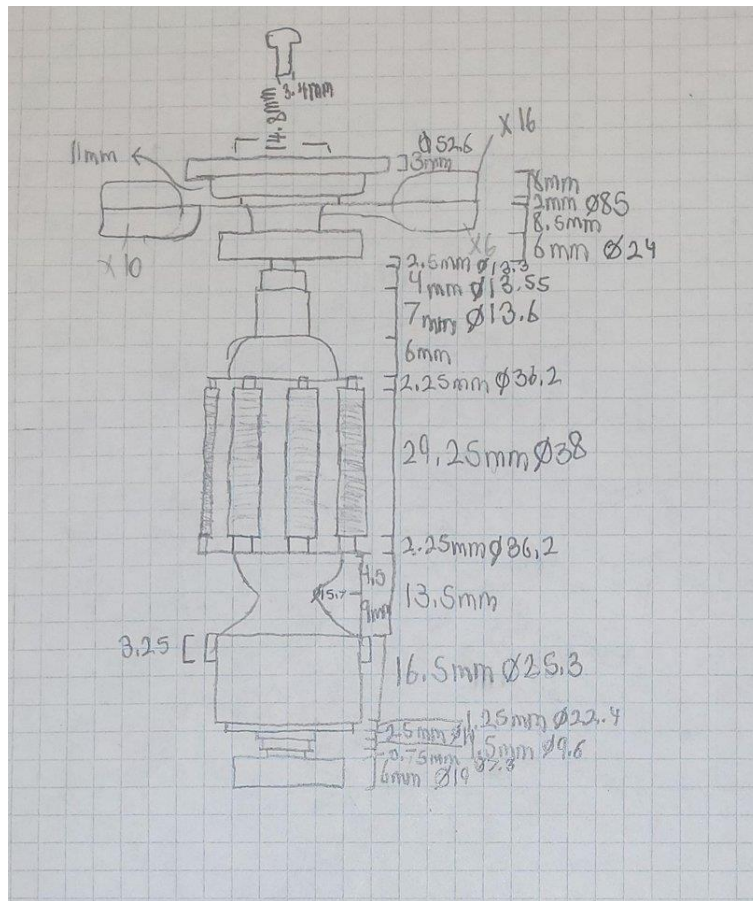


Figure 3. Fusion 360 model of the sander backing pad showing the central hub mount and radial dust-collection port pattern.



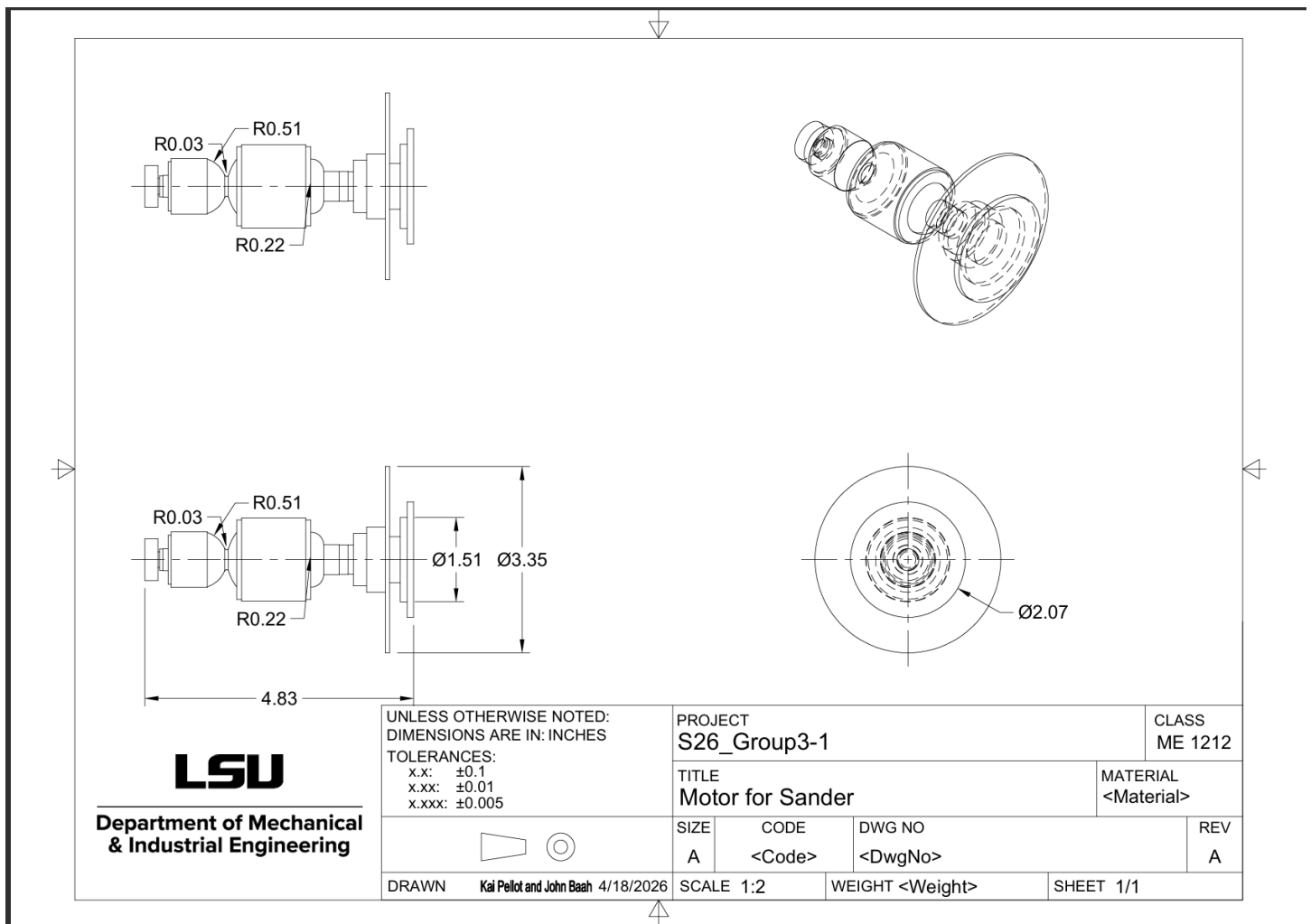


Figure 4. Hand sketch — motor shaft and drivetrain component stack with measured dimensions (all dimensions in mm).

Fit

The fit of the drivetrain is composed of tight-tolerance interactions that prevent slippage between components. The ball bearings at both ends of the shaft are installed via interference fit — pressed onto the shaft with high precision to ensure zero relative motion. The upper bearing (Ø13.6 mm bore, 7 mm height) and the lower bearing (Ø22.4 mm bore, 1.25 mm section) are each retained by the housing. The metal counterweight uses a keyed fit, interlocking with the shaft to synchronize rotation. The eccentric hub is tightly fit onto the lower shaft section (Ø25.3 mm) and positioned offset from the shaft centerline, driving the orbital motion of the backing pad. As shown in the dimensioned drawing (Figure 5), the backing pad's central bore (Ø42 mm) fits over the eccentric hub post, while the eight evenly spaced dust holes (Ø10 mm on a Ø108.75 mm bolt circle at 45° intervals) align with the dust-collection channels in the lower housing at all rotational positions.

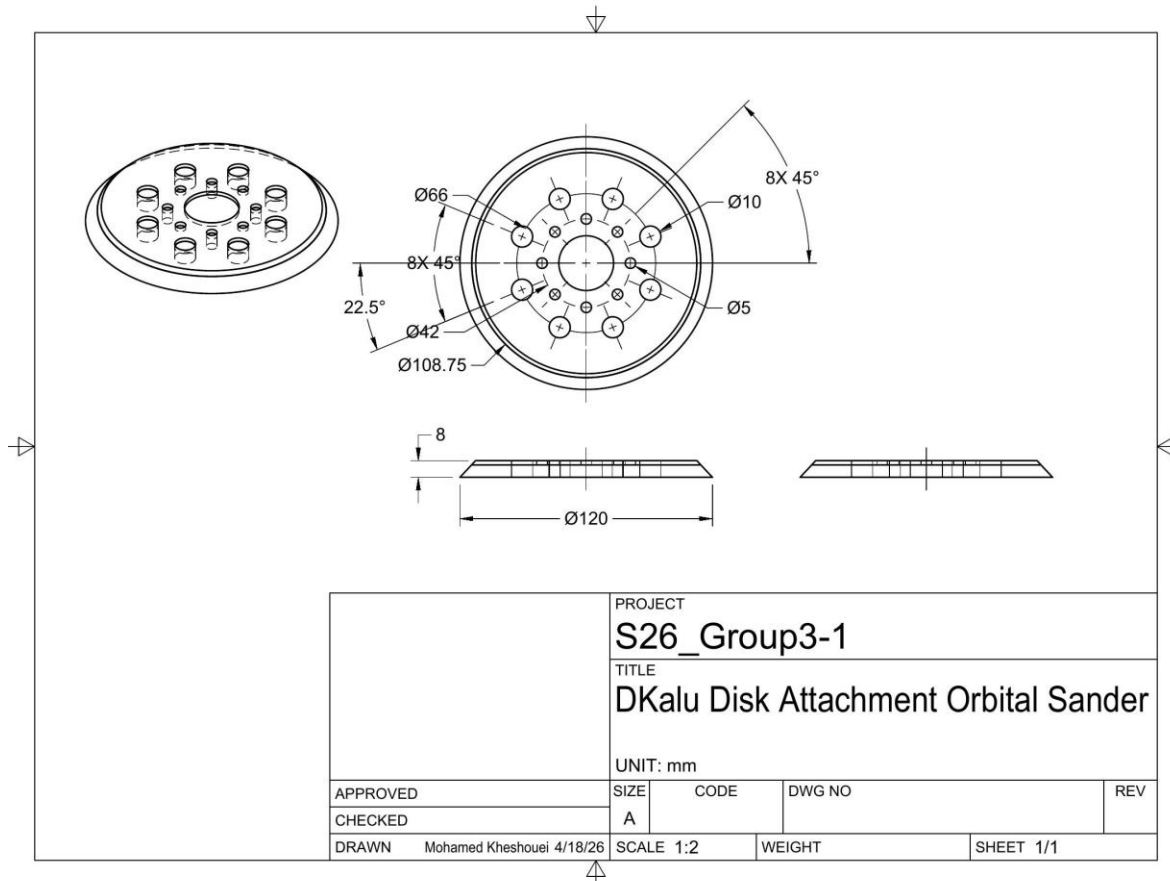


Figure 5. Dimensioned drawing — Disk Attachment, Orbital Sander (drawn by D. Kalu, Group 3-1). Ø120 mm overall; Ø42 mm central bore; 8 × Ø10 mm dust holes on Ø108.75 mm PCD at 45° intervals.

Function

The primary function of the drivetrain is to convert rotational energy from the motor into the controlled circular orbit that defines the sander's sanding action. The motor spins the shaft, causing the eccentric hub to drive the backing pad in rapid, small overlapping circles. The counterweight offsets the centrifugal force generated by the eccentric hub, preventing uncontrollable vibration and enabling smooth, precise sanding. The backing pad transmits orbital motion directly to the sandpaper and work surface. The radial hole pattern in the backing pad also functions as the intake manifold for the dust-collection circuit described in Section 4. Collectively, these components fulfill the sub-functions of generating rotational motion and interlinking with the abrasive attachment as identified in the functional decomposition tree (Figure 1).

3.2 Variations / Improvements(DK)

One weakness of the current drivetrain system is the intense combined load placed on the lower ball bearing of the motor shaft. This bearing sits in close proximity to both the motor windings and the eccentric hub — two components that generate significant heat during operation. While the internal fan assists with cooling, the bearing's lubricating grease degrades over time under sustained thermal exposure. Additionally, the eccentric hub continuously applies lateral radial forces on this bearing, accelerating wear beyond what a purely axial load would produce. Progressive failure of this bearing leads to shaft misalignment, increased vibration, and ultimately total mechanical failure of the device.

The proposed improvement is to replace the current steel ball bearings with ceramic hybrid bearings (steel races with silicon nitride ceramic balls). Ceramic balls expand significantly less than steel under thermal stress, reducing internal clearance changes that accelerate wear. They exhibit a lower coefficient of friction and can withstand substantially higher operating temperatures — directly addressing the thermal vulnerability of the lower bearing position. Ceramic is also harder than steel, increasing resistance to the radial loads imposed by the eccentric hub.

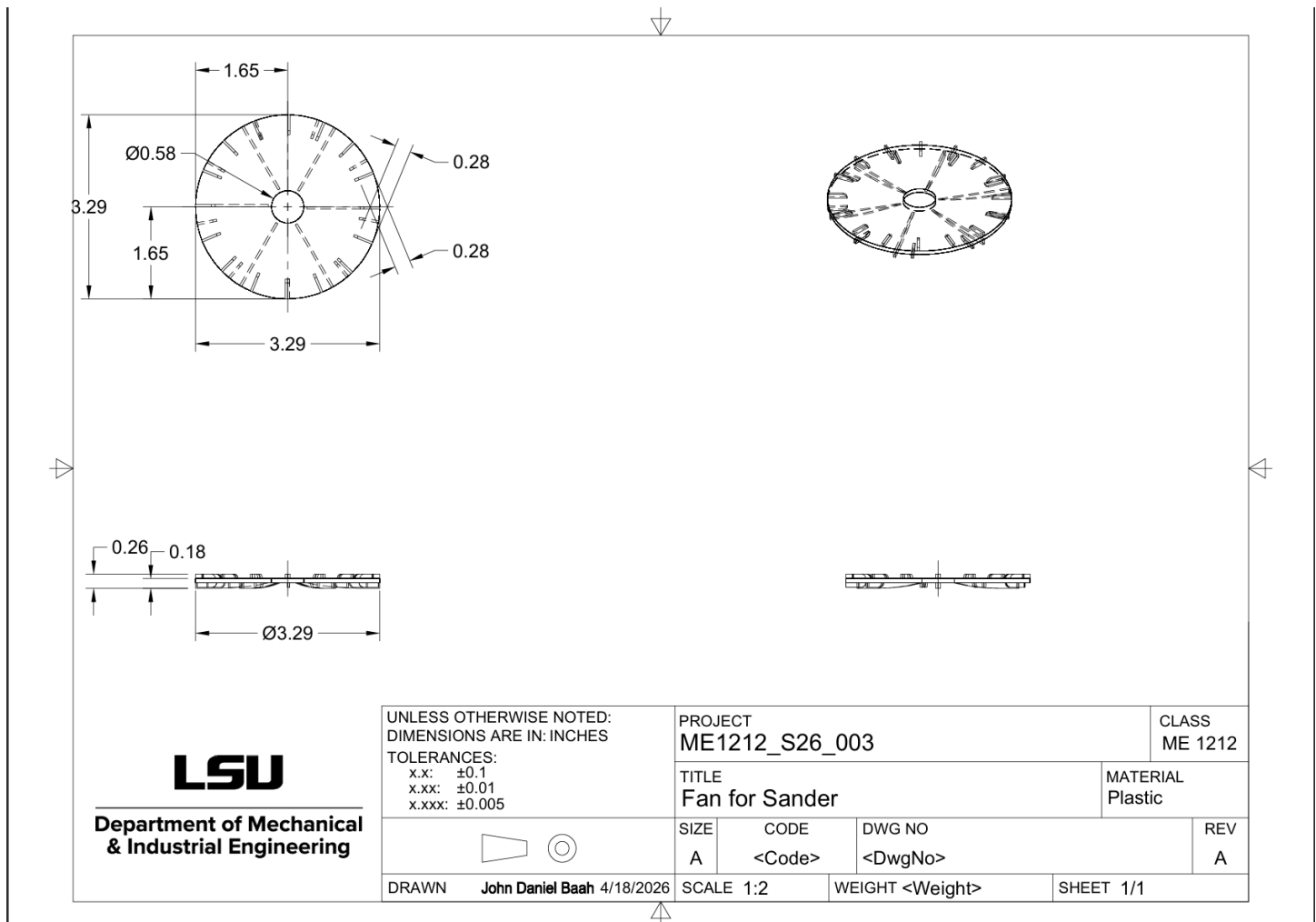
This substitution would increase per-unit manufacturing cost, as ceramic bearings carry a price premium over steel. However, ceramic bearings typically last two to five times longer than steel equivalents under comparable operating conditions, reducing replacement frequency and lowering long-term cost of ownership. From a user experience perspective, the sander would run cooler, produce less operational noise, and have a meaningfully longer usable lifespan.

4. Thermal/Fluid System(KP,JDSB)

4.1 Form, Fit, and Function

Form

The Thermal/Fluid system consists of the internal cooling fan, the motor housing vents, the dust-collection port, and the dust bag attachment. The cooling fan assembly, documented in the hand sketch (Figure 6), is a circular disc ($\varnothing 83.57$ mm overall, $\varnothing 51.57$ mm inner ring) mounted directly on top of the motor shaft. The fan carries 16 rectangular blade slots arranged radially around the inner ring and a ring of 6 small circular holes ($\varnothing 6$ mm) positioned at a pitch circle diameter of approximately $\varnothing 46$ mm. The fan thickness is approximately 15.67 mm as noted on the sketch. The housing, documented in the dimensioned drawing (Figure 7), provides the structural enclosure for this airflow circuit, with an overall length of 172.2 mm, width of 143.34 mm, and height of 116 mm. Molded ventilation slots on the sides of the housing (R44.61 mm vent curve radius) provide intake and exhaust pathways for cooling airflow. At the base, the backing pad's radial through-holes connect to the lower housing dust channels that route collected debris to the external dust bag port.



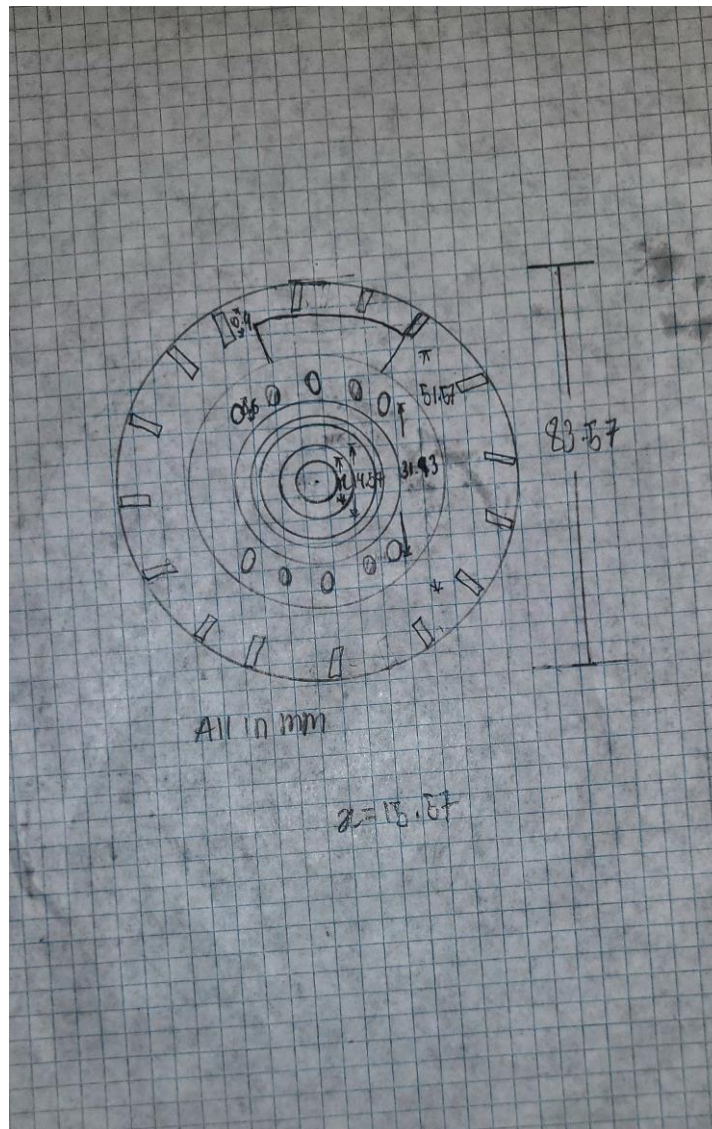


Figure 6. Hand sketch — fan and counterweight assembly with measured dimensions (all dimensions in mm).

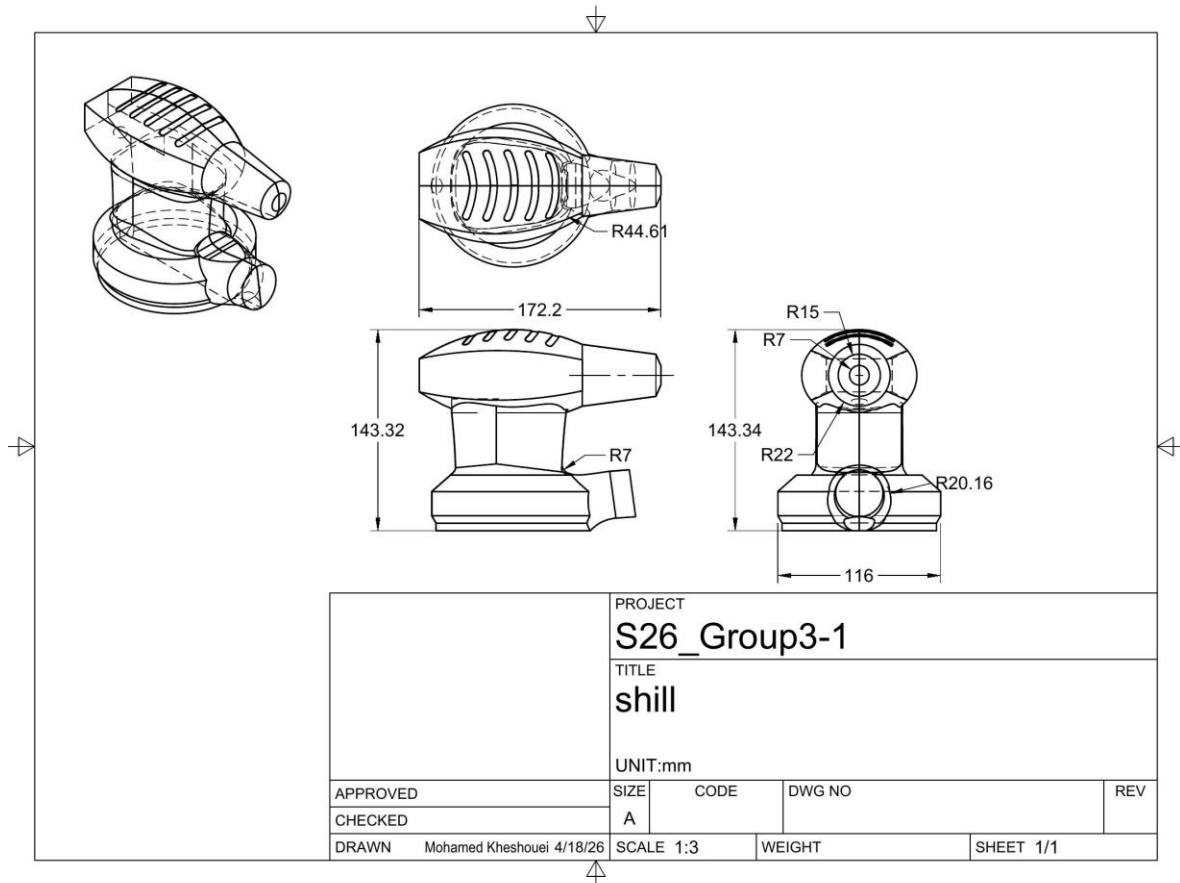


Figure 7. Dimensioned drawing — Sander Housing (drawn by M. Kheshouei, Group 3-1). 172.2 × 143.34 × 116 mm overall; R44.61 mm vent curve; R22 and R20.16 mm grip transitions.

Fit

The cooling fan is press-fit directly onto the motor shaft, rotating in synchrony with it without a separate coupling. The fan blade geometry is sized to fit within a tight cylindrical clearance inside the upper housing, maximizing airflow velocity through the vent slots without contact. The 6 small holes in the fan disc (Figure 6) align with the motor shaft mounting features and assist in securing the fan axially. The dust port in the lower housing aligns precisely with the outlet channel molded into the housing body, joined during assembly via a snap-fit or adhesive bond. The dust bag couples to the external port via a friction-fit rubber collar, allowing tool-free removal for emptying. The through-hole pattern in the backing pad is distributed radially so that dust intake channels remain aligned with the housing inlet ports at all rotational positions of the pad.

Function

The Thermal/Fluid system serves two integrated purposes: cooling the motor and collecting sanding debris. As the motor shaft rotates, the fan draws ambient air in through the side vents of the housing, passes it over the motor windings to dissipate heat, and exhausts it through the opposite vent set. This continuous airflow prevents

heat buildup in the windings and bearings, extending motor service life. Simultaneously, the orbital motion of the backing pad draws dust and fine particles up through the pad's hole pattern, through the lower housing channel, and out through the dust port into the dust bag. This integrated dust collection reduces airborne particle exposure for the user and keeps the work surface clear. Both functions correspond directly to the Waste Collection and thermal regulation sub-functions in the functional decomposition tree (Figure 1).

4.2 Variations / Improvements

A notable weakness of the current Thermal/Fluid system is the limited effectiveness of the passive dust collection design. The fabric dust bag captures larger particles but allows fine dust — including sub-10-micron particles that pose the greatest respiratory health risk — to pass through and re-enter the air. The bag also requires manual detachment and frequent emptying during use, interrupting workflow. When overfilled, collection efficiency drops sharply, increasing back-pressure that reduces both dust capture effectiveness and cooling airflow to the motor.

The proposed improvement is to replace the fabric dust bag with a compact two-stage system: a cyclonic pre-separator feeding a sealed HEPA-rated filter canister. A cyclonic separator uses centrifugal force to spin heavier particles to the outer wall and drop them into a collection chamber before air reaches the filter, dramatically reducing filter load. The remaining airstream then passes through a HEPA filter element, capturing particles as small as 0.3 microns at 99.97% efficiency — well above the performance of the current fabric bag.

This upgrade would increase manufacturing cost due to the additional components and tighter tolerances required for the cyclonic chamber and sealed filter housing. However, the HEPA filter's service life would be substantially extended by the pre-separator, keeping long-term consumable costs manageable. The sander's own lifespan would benefit from reduced fine-dust ingress into the motor and bearing cavity. Most importantly, the improvement would significantly reduce the user's exposure to hazardous fine particulate matter, a meaningful health benefit especially during extended use.

5. Human Interfaces System (MK, DAW)

5.1 Form, Fit, and Function

Form

The Human Interfaces system encompasses every component through which the user physically interacts with the orbital sander: the outer housing grip body, the on/off trigger switch, the power cord, and the strain relief boot. The outer housing is a two-piece injection-molded plastic shell that forms the ergonomic palm-grip shape of the sander. As documented in the housing dimensioned drawing (Figure 7), the housing measures 172.2 mm in length and tapers from 143.34 mm wide at the motor section down to a narrower grip neck, with smooth curved transitions (R22 mm and R20.16 mm radii) that conform to the natural contour of a user's hand. The top surface of the grip features a series of parallel longitudinal ridges molded into the plastic to improve traction during use. A rocker-style on/off switch is recessed into the front face of the upper housing, accessible by the user's thumb or index finger during normal operation. The power cord exits the rear of the housing through a flexible rubber strain relief boot measuring approximately 11 mm in outer diameter at the housing entry point.

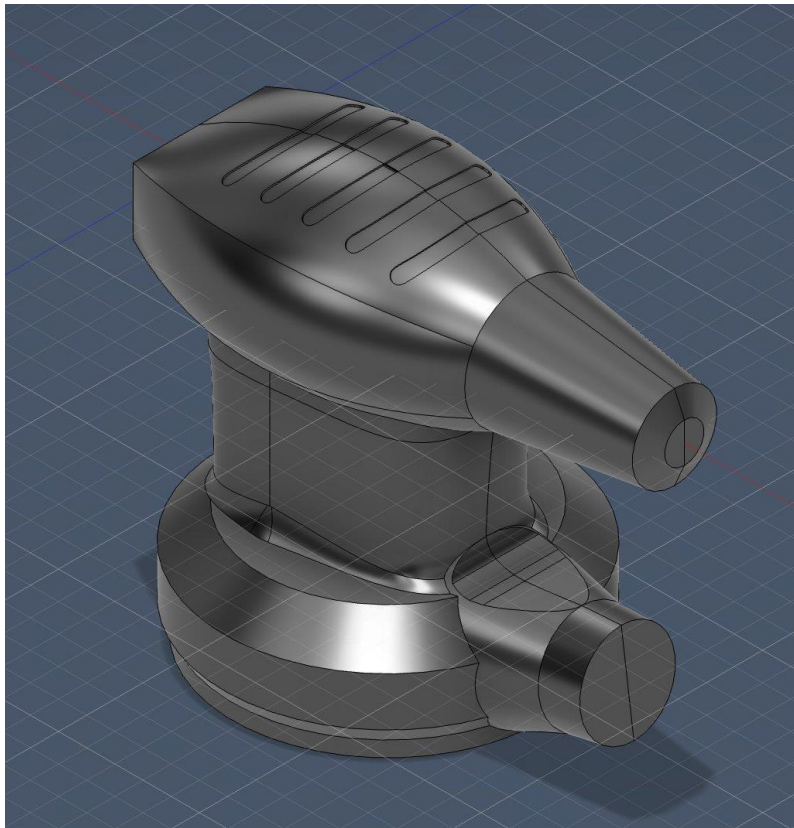


Figure 2. Fusion 360 Assembly — Full Orbital Sander, showing overall drivetrain housing geometry.

Fit

The two halves of the outer housing shell are joined by self-tapping plastic screws recessed into the lower half, threading into molded bosses in the upper half to create a rigid, flush seam. The on/off switch is a snap-in component that clips into a molded rectangular cutout in the front face of the housing, held captive by integral locking tabs, and connects to the internal wiring via push-on crimp terminals. The strain relief boot is a separate elastomeric component press-fit into the cord exit hole at the rear of the housing, clamping the outer jacket of the power cord through a tapered interference fit that resists pull-out forces. The grip ridges on the top surface are co-molded with the housing body, keyed to the surface geometry to prevent the rubber overmold from shifting or peeling under sustained hand pressure.

Function

The Human Interfaces system serves as the ergonomics and control layer between the user and the machine. The shaped housing allows the user to grip and guide the sander comfortably during extended use, with grip ridges and the tapered palm form reducing hand fatigue and slippage. The on/off switch provides simple, reliable power control — a single press engages or disengages the motor circuit — with its recessed position preventing accidental activation during handling. The power cord delivers AC electrical power to the motor, while the strain relief boot protects the cord from fatigue cracking at the high-stress entry point into the housing. Together, these components fulfill the Control/Regulate inputted energy function identified in the functional decomposition tree (Figure 1) and represent the primary physical interface through which user intent is translated into device operation.

5.2 Variations / Improvements

A key weakness in the Human Interfaces system is the rocker switch, which provides limited discrete speed control (levels 1–6) rather than continuous adjustment. While this allows for some variation in motor speed, the fixed settings may still be too coarse for delicate applications. Users working on sensitive surfaces such as thin wood veneers, automotive paint, or finish coats may find that even the lowest setting is too aggressive for fine finishing work. This limitation reduces the versatility of the tool and increases the risk of surface damage.

The proposed improvement is the addition of a variable-speed thumbwheel dial integrated into the rear grip of the housing, paired with a pulse-width modulation (PWM) electronic speed controller (ESC) circuit board housed inside the upper shell. The thumbwheel would allow the user to continuously adjust motor speed from a low setting suitable for fine polishing up to full speed for aggressive material removal. PWM speed control is a well-established, cost-effective technique widely used in handheld power tools and is compatible with the existing motor type.

Adding variable speed control would increase the bill of materials cost due to the ESC circuit board and a more complex control mechanism. However, the added functionality substantially expands the range of use cases for the sander, improving its overall value. From a lifespan perspective, operating the motor at lower speeds when full power is unnecessary reduces thermal and mechanical stress, potentially extending motor and bearing life. The improvement in user experience is significant: users gain precise control over the sanding process, reducing the likelihood of surface damage and making the tool suitable for a broader range of professional and hobbyist applications.

6. Project Management (MK)

Team 3-1 managed this project using a detailed Gantt chart (Figure 8) that tracked all major tasks from the initial team contract through the final assembly and animation deliverable. The project was divided into group-wide tasks and system-specific tasks, with the Drivetrain, Human Interfaces, and Thermal/Fluid systems each assigned to a dedicated pair of team members, enabling parallel progress on CAD modeling, form-fit-function writing, and improvement research. Lab sessions from March 18 through April 8 were used for disassembly, documentation, and measurement, while CAD modeling, technical writing, and research were conducted outside of class. Key milestones — including the Functional Decomposition submission (March 29), the Interim Gantt Chart (April 2), and this Technical Report (April 19) — were identified early and built into the schedule. Report drafting and review were allocated April 13–18, providing time for revision prior to submission. The Gantt chart was updated throughout the project to reflect actual completion dates and any adjustments made in response to lab constraints or modeling complexity.

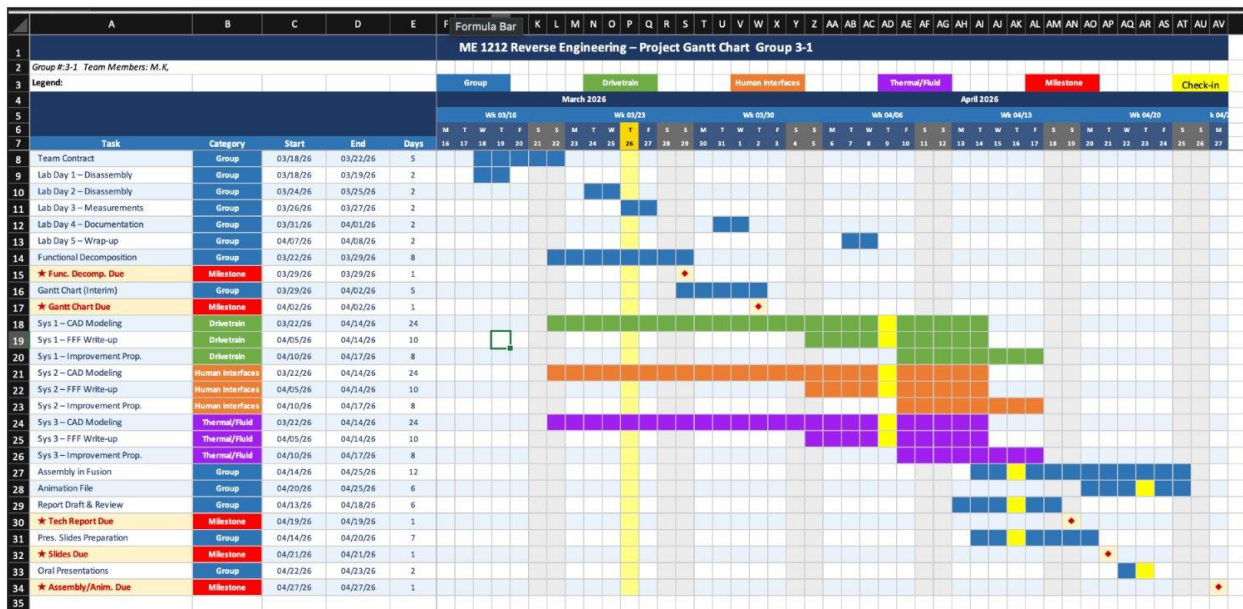


Figure 8. Project Gantt Chart — ME 1212 Reverse Engineering Project, Group 3-1 (updated April 2026).